In [ ]:
```python
## Anandhan - 2025166043 - DSLab Mock Test - 20/11/2025

import numpy as np
import pandas as pd
import seaborn as sns
import os
from scipy import stats
import matplotlib.pyplot as plt

file_path = "BookDataset.csv"
data = pd.read_csv(file_path)

df = pd.DataFrame(data)
print(df.head())
```

```
       Company   body-style  wheel-base  length  horsepower  average-mileage  \
0  alfa-romero  convertible        88.6   168.8         111               21
1  alfa-romero  convertible        88.6   168.8         111               21
2  alfa-romero    hatchback        94.5   171.2         154               19
3         audi        Sedan        99.8   176.6         102               24
4         audi        Sedan        99.4   176.6         115               18

   price
0  13495
1  16500
2  16500
3  13950
4  17450
```

In [2]:
```python
print("Shape of the dataset\n", df.shape)
```

```
Shape of the dataset
 (42, 7)
```

In [4]:
```python
print("First Five Rows \n", df.head(5))
print("Last Five Rows\n", df.tail(5))
```

```
First Five Rows
        Company   body-style  wheel-base  length  horsepower  average-mileage  \
0  alfa-romero  convertible        88.6   168.8         111               21
1  alfa-romero  convertible        88.6   168.8         111               21
2  alfa-romero    hatchback        94.5   171.2         154               19
3         audi        Sedan        99.8   176.6         102               24
4         audi        Sedan        99.4   176.6         115               18

   price
0  13495
1  16500
2  16500
3  13950
4  17450
Last Five Rows
     Company   body-style  wheel-base  length  horsepower  average-mileage  \
37    nissan        sedan        94.5   165.3          55               45
38    nissan        wagon        94.5   170.2          69               31
39    nissan        sedan       100.4   184.6         152               19
40   porsche      hardtop        89.5   168.9         207               17
41   porsche  convertible        89.5   168.9         207               17

    price
37   7099
38   7349
39  13499
40  34028
41  37028
```

In [34]:

```
print("The Most Expensive Car Company Name: ", df.loc[df['price'].idxmax()]["Com
```

```
The Most Expensive Car Company Name:  mercedes-benz
```

In [35]:

```
print(df.isnull().sum())
cleaned_dataset = df.dropna()
cleaned_dataset.to_csv("Cleaned_DataSet.csv", index = False)
#pd.to_csv(cleaned_dataset)
```

```
Company            0
body-style         0
wheel-base         0
length             0
horsepower         0
average-mileage    0
price              0
capacity           0
dtype: int64
```

In [11]:

```
df["capacity"] = df["horsepower"]/df["average-mileage"]
print(df["capacity"])
```

```
0      5.285714
1      5.285714
2      8.105263
3      4.250000
4      6.388889
5      5.789474
6      5.789474
7      4.391304
8      4.391304
9      5.761905
10    11.375000
11    11.375000
12    12.133333
13     1.021277
14     1.842105
15     1.842105
16     2.193548
17     2.193548
18     2.533333
19     4.208333
20     4.000000
21     3.250000
22    11.733333
23    11.733333
24    20.153846
25     2.266667
26     2.193548
27     2.193548
28     5.941176
29     2.322581
30     5.590909
31    13.142857
32    13.142857
33     1.837838
34     2.193548
35     3.520000
36     3.520000
37     1.222222
38     2.225806
39     8.000000
40    12.176471
41    12.176471
Name: capacity, dtype: float64
```

In [12]:

```
print(df[df.Company == "bmw"])
```

```
  Company body-style  wheel-base  length  horsepower  average-mileage  price  \
7       bmw      sedan       101.2   176.8         101               23  16430
8       bmw      sedan       101.2   176.8         101               23  16925
9       bmw      sedan       101.2   176.8         121               21  20970
10      bmw      sedan       103.5   189.0         182               16  30760
11      bmw      sedan       103.5   193.8         182               16  41315
12      bmw      sedan       110.0   197.0         182               15  36880

     capacity
7    4.391304
8    4.391304
9    5.761905
10  11.375000
11  11.375000
12  12.133333
```

In [37]:
```
print("Total cars per company: ", df['Company'].value_counts())
```

```
Total cars per company:  Company
bmw              6
mazda            5
audi             4
mitsubishi       4
alfa-romero      3
nissan           3
chevrolet        3
jaguar           3
honda            3
mercedes-benz    3
dodge            2
porsche          2
isuzu            1
Name: count, dtype: int64
```

In [38]:
```
print("Each company highest price cars", df.groupby("Company")["price"].max())
```

```
Each company highest price cars Company
alfa-romero      16500
audi             18920
bmw              41315
chevrolet         6575
dodge             6377
honda            12945
isuzu             6785
jaguar           36000
mazda            18344
mercedes-benz    45400
mitsubishi        8189
nissan           13499
porsche          37028
Name: price, dtype: int64
```

In [41]:
```
print("Average mileage for each car company", df.groupby("Company")["average-mil
```

```
Average mileage for each car company Company
alfa-romero      20.333333
audi             20.000000
bmw              19.000000
chevrolet        41.000000
dodge            31.000000
honda            26.333333
isuzu            24.000000
jaguar           14.333333
mazda            28.000000
mercedes-benz    16.666667
mitsubishi       29.500000
nissan           31.666667
porsche          17.000000
Name: average-mileage, dtype: float64
```

In [42]:

```
print("Sorted Cars by Price", df.sort_values(by = "price"))
```

Sorted Cars by Price

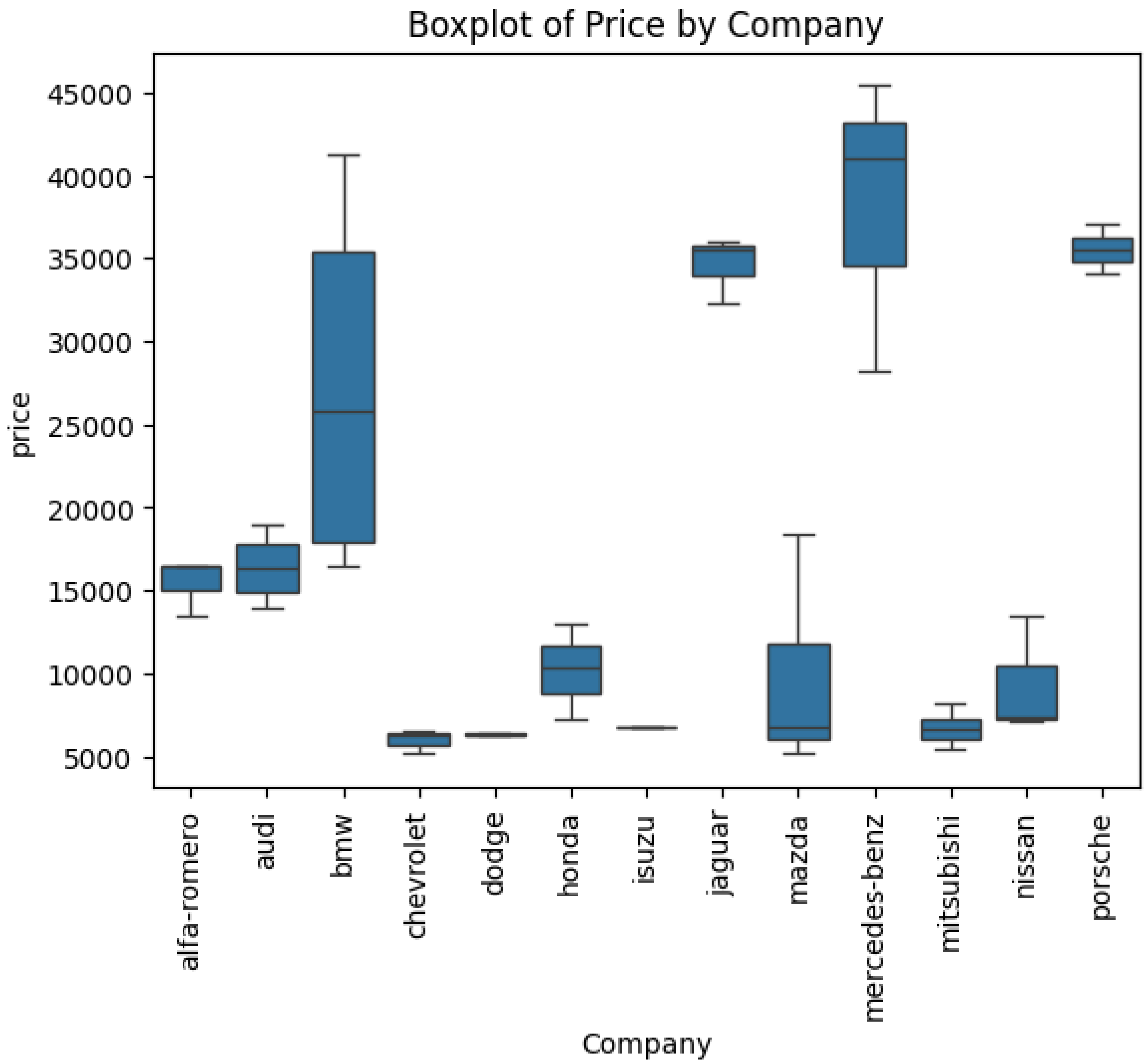
	Company	body-style	wheel-base	length	horsepower
13	chevrolet	hatchback	88.4	141.1	48
25	mazda	hatchback	93.1	159.1	68
33	mitsubishi	hatchback	93.7	157.3	68
26	mazda	hatchback	93.1	159.1	68
34	mitsubishi	hatchback	93.7	157.3	68
17	dodge	hatchback	93.7	157.3	68
14	chevrolet	hatchback	94.5	155.9	70
16	dodge	hatchback	93.7	157.3	68
15	chevrolet	sedan	94.5	158.8	70
21	isuzu	sedan	94.3	170.7	78
27	mazda	hatchback	93.1	159.1	68
35	mitsubishi	sedan	96.3	172.4	88
37	nissan	sedan	94.5	165.3	55
18	honda	wagon	96.5	157.1	76
38	nissan	wagon	94.5	170.2	69
36	mitsubishi	sedan	96.3	172.4	88
20	honda	sedan	96.5	169.1	100
28	mazda	hatchback	95.3	169.0	101
19	honda	sedan	96.5	175.4	101
0	alfa-romero	convertible	88.6	168.8	111
39	nissan	sedan	100.4	184.6	152
3	audi	Sedan	99.8	176.6	102
5	audi	sedan	99.8	177.3	110
7	bmw	sedan	101.2	176.8	101
1	alfa-romero	convertible	88.6	168.8	111
2	alfa-romero	hatchback	94.5	171.2	154
8	bmw	sedan	101.2	176.8	101
4	audi	Sedan	99.4	176.6	115
29	mazda	sedan	104.9	175.0	72
6	audi	wagon	105.8	192.7	110
9	bmw	sedan	101.2	176.8	121
30	mercedes-benz	wagon	110.0	190.9	123
10	bmw	sedan	103.5	189.0	182
22	jaguar	sedan	113.0	199.6	176
40	porsche	hardtop	89.5	168.9	207
23	jaguar	sedan	113.0	199.6	176
24	jaguar	sedan	102.0	191.7	262
12	bmw	sedan	110.0	197.0	182
41	porsche	convertible	89.5	168.9	207
31	mercedes-benz	sedan	120.9	208.1	184
11	bmw	sedan	103.5	193.8	182
32	mercedes-benz	hardtop	112.0	199.2	184

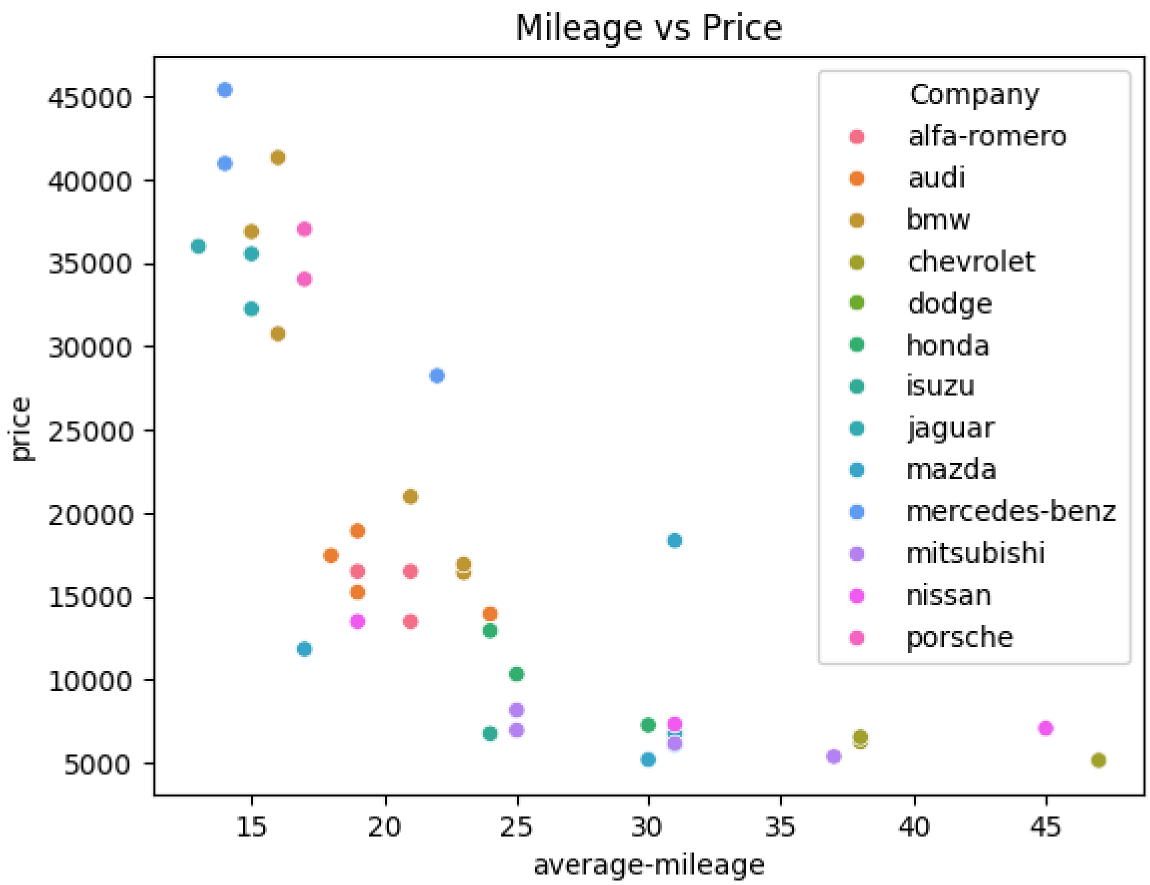
	average-mileage	price	capacity
13	47	5151	1.021277
25	30	5195	2.266667
33	37	5389	1.837838
26	31	6095	2.193548
34	31	6189	2.193548
17	31	6229	2.193548
14	38	6295	1.842105
16	31	6377	2.193548
15	38	6575	1.842105
21	24	6785	3.250000
27	31	6795	2.193548
35	25	6989	3.520000
37	45	7099	1.222222
18	30	7295	2.533333

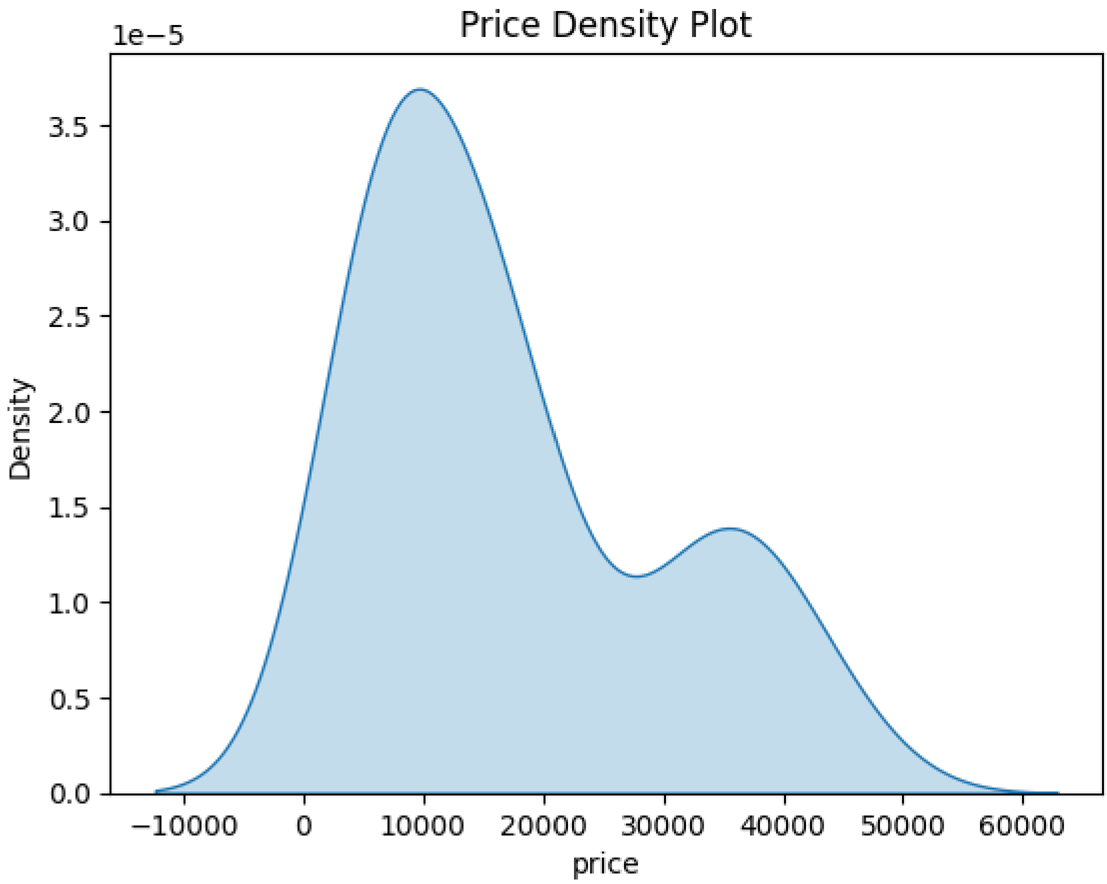
38	31	7349	2.225806
36	25	8189	3.520000
20	25	10345	4.000000
28	17	11845	5.941176
19	24	12945	4.208333
0	21	13495	5.285714
39	19	13499	8.000000
3	24	13950	4.250000
5	19	15250	5.789474
7	23	16430	4.391304
1	21	16500	5.285714
2	19	16500	8.105263
8	23	16925	4.391304
4	18	17450	6.388889
29	31	18344	2.322581
6	19	18920	5.789474
9	21	20970	5.761905
30	22	28248	5.590909
10	16	30760	11.375000
22	15	32250	11.733333
40	17	34028	12.176471
23	15	35550	11.733333
24	13	36000	20.153846
12	15	36880	12.133333
41	17	37028	12.176471
31	14	40960	13.142857
11	16	41315	11.375000
32	14	45400	13.142857

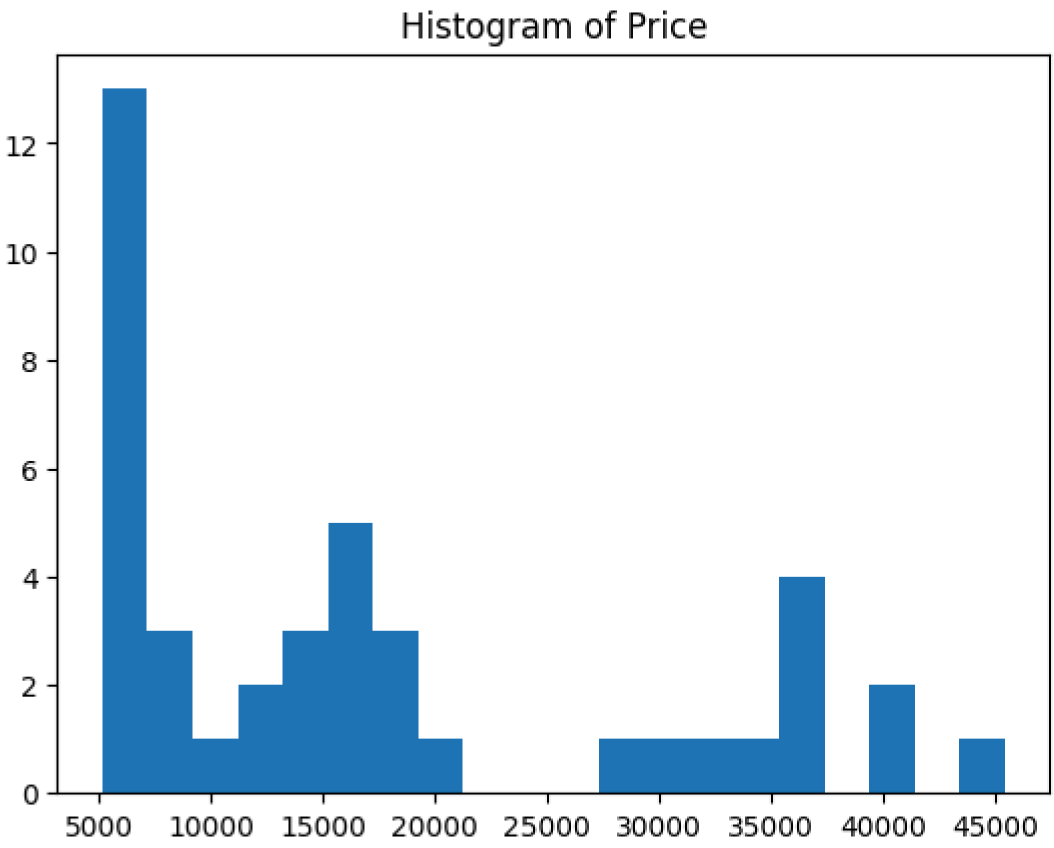
In [50]:

```
sns.boxplot(x="Company", y="price", data=df)
plt.xticks(rotation=90)
plt.title("Boxplot of Price by Company")
plt.show()

sns.scatterplot(x="average-mileage", y="price", hue="Company", data=df)
plt.title("Mileage vs Price")
plt.show()

sns.kdeplot(df["price"], fill=True)
plt.title("Price Density Plot")
plt.show()

plt.hist(df["price"], bins=20)
plt.title("Histogram of Price")
plt.show()
```


Boxplot of Price by Company
price
45000
40000
35000
30000
25000
20000
15000
10000
5000
alfa-romero
audi
bmw
chevrolet
dodge
honda
isuzu
jaguar
mazda
mercedes-benz
mitsubishi
nissan
porsche
Company


Mileage vs Price
price
45000
40000
35000
30000
25000
20000
15000
10000
5000
15
20
25
30
35
40
45
average-mileage
Company
alfa-romero
audi
bmw
chevrolet
dodge
honda
isuzu
jaguar
mazda
mercedes-benz
mitsubishi
nissan
porsche





In [46]:

```
mct = df.groupby("Company")["horsepower"].agg(["mean", "median", lambda x: x.mod
print(mct)
print("Mean: ", df["horsepower"].mean())
print("Median: ", df["horsepower"].median())
print("Mode: ", df["horsepower"].mode())
```

```
                    mean  median  <lambda_0>
Company
alfa-romero    125.333333   111.0         111
audi           109.250000   110.0         110
bmw            144.833333   151.5         182
chevrolet       62.666667    70.0          70
dodge           68.000000    68.0          68
honda           92.333333   100.0          76
isuzu           78.000000    78.0          78
jaguar         204.666667   176.0         176
mazda           75.400000    68.0          68
mercedes-benz  163.666667   184.0         184
mitsubishi      78.000000    78.0          68
nissan          92.000000    69.0          55
porsche        207.000000   207.0         207
Mean:  115.35714285714286
Median:  101.0
Mode:  0    68
Name: horsepower, dtype: int64
```

In [29]:
```
print("Covariance", df[["average-mileage", "price"]].cov())
print("Correlation", df[["average-mileage", "price"]].corr())
```

```
Covariance                  average-mileage         price
average-mileage        71.991289 -7.975121e+04
price              -79751.210801  1.517337e+08
Correlation                  average-mileage     price
average-mileage         1.000000 -0.763056
price                  -0.763056  1.000000
```

In [48]:
```
matrix = df[["horsepower", "average-mileage", "price"]].corr()
eigen_values, eigen_vectors = np.linalg.eig(matrix)
print("Eigen Value", eigen_values)
print("Eigen Vectors", eigen_vectors)
```

```
Eigen Value [2.66390795 0.08546474 0.25062731]
Eigen Vectors [[ 0.59275565 -0.77247652  0.22786128]
 [-0.55948971 -0.19144647  0.80642391]
 [ 0.5793203   0.60549837  0.54567364]]
```